

Dynamic Guideline: WHO-Based Tuberculosis - Diagnosis and/or Treatment and/or Management

Evidence integrity note: This report is written in English and uses WHO tuberculosis guideline documents as the primary source. The prompt requests a most recent guideline titled “**Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline.**” This appears to describe a **WHO guideline-development case study**, not a clinical tuberculosis guideline. It is therefore treated as a methodological document linked to youth engagement—most relevant to WHO child/adolescent TB guideline work—but **not counted as one of the five clinical WHO tuberculosis guidelines** synthesized below.

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO guideline used as anchor document

- **Guideline Title:** *WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities, 2nd ed.*
 - **Publication / latest-update date used for this dynamic report:** 29 April 2026
 - **Target Population:**
 - Adults, adolescents, children, and pregnant or breastfeeding people with tuberculosis or at risk of tuberculosis who have comorbid conditions.
 - Priority groups include people living with HIV, people with diabetes mellitus, undernutrition, harmful alcohol or substance use, viral hepatitis risk, chronic lung disease, mental health conditions, and other conditions affecting TB outcomes.
 - **Scope and Objectives:**
 - Provide consolidated WHO recommendations on the diagnosis, treatment, prevention, and integrated management of TB with comorbidities.
 - Promote patient-centred, integrated care to reduce mortality, transmission, treatment failure, relapse, drug toxicity, and loss to follow-up.
 - Align TB services with HIV, diabetes, nutrition, mental health, substance-use, primary-care, and social-support programmes.
-

1.2 WHO Guideline Search Scope: Exactly 5 WHO Tuberculosis Guidelines Synthesized

No.	WHO Guideline	Clinical Area	Included in This Report
-----	---------------	---------------	-------------------------

1 <i>WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities, 2nd ed.</i>	TB and comorbidities	Yes
2 <i>WHO consolidated guidelines on tuberculosis: module 3: diagnosis: rapid diagnostics for tuberculosis detection</i>	Diagnosis and drug-susceptibility testing	Yes
3 <i>WHO consolidated guidelines on tuberculosis: module 4: treatment: drug-susceptible tuberculosis treatment</i>	Drug-susceptible TB treatment	Yes
4 <i>WHO consolidated guidelines on tuberculosis: module 4: treatment: drug-resistant tuberculosis treatment, 2022 update</i>	MDR/RR-TB and pre-XDR/XDR-TB treatment	Yes
5 <i>WHO consolidated guidelines on tuberculosis: module 5: management of tuberculosis in children and adolescents</i>	Paediatric and adolescent TB	Yes

1.3 Key WHO Recommendations With GRADE Ratings

A. TB and HIV, Comorbidities, and Integrated Care

Recommendation	GRADE Evidence Quality	Strength
All people with TB should be offered HIV testing, and all people living with HIV should be systematically screened for TB at each contact with health services.	High	Strong
Antiretroviral therapy should be started as soon as possible and generally within 2 weeks of starting TB treatment in people with HIV-associated TB, except where delayed initiation is clinically indicated, such as TB meningitis.	High	Strong
Co-trimoxazole preventive therapy should be provided to people living with HIV with active TB disease according to WHO HIV guidance.	High	Strong
TB preventive treatment should be offered to eligible people living with HIV after active TB disease has been excluded.	High	Strong
WHO-recommended rapid molecular tests should be used as the initial diagnostic test for TB and rifampicin resistance in people with signs or symptoms of TB, including people living with HIV.	High	Strong
Lateral-flow urine lipoarabinomannan testing should be used to assist TB diagnosis in seriously ill, hospitalized, or advanced-HIV patients according to WHO eligibility criteria.	Moderate	Strong
People with TB should be assessed for diabetes mellitus in settings where diabetes prevalence is high or where individual risk factors are present.	Low	Conditional
People with diabetes should be screened for TB symptoms and evaluated for TB when symptomatic or at increased risk.	Low	Conditional
All people with TB should receive nutritional assessment, counselling, and support; severe acute malnutrition should be managed urgently according to WHO nutrition guidance.	Moderate	Strong

TB programmes should identify and address harmful alcohol use, substance use, depression, anxiety, and other mental health conditions that impair adherence and outcomes.	Low	Conditional
TB care should be integrated with HIV, diabetes, nutrition, primary-care, mental-health, and social-protection services where feasible.	Moderate	Strong

B. TB Diagnosis and Drug-Susceptibility Testing

Recommendation	GRADE Evidence Quality	Strength
WHO-recommended rapid molecular diagnostic tests, such as Xpert MTB/RIF Ultra or other WHO-endorsed nucleic-acid amplification tests, should be used as the initial diagnostic test for pulmonary TB rather than smear microscopy alone.	High	Strong
Rapid molecular testing should be used to detect rifampicin resistance at the time of initial TB diagnosis.	High	Strong
In people at risk of drug-resistant TB, rapid molecular tests for fluoroquinolone resistance should be used to guide regimen selection.	Moderate	Strong
Culture and phenotypic drug-susceptibility testing remain important for confirmation, expanded resistance testing, and management of complex drug-resistant TB.	Moderate	Strong
For children and people unable to produce sputum, WHO-recommended non-sputum specimens, such as gastric aspirate, stool, nasopharyngeal aspirate, or other validated samples, may be used with molecular tests where available.	Moderate	Conditional
LF-LAM should be used as an adjunct diagnostic test in eligible people living with HIV, especially hospitalized or seriously ill patients.	Moderate	Strong

C. Drug-Susceptible TB Treatment

Recommendation	GRADE Evidence Quality	Strength
Standard 6-month rifampicin-based therapy remains an effective regimen for most people with drug-susceptible pulmonary TB: 2 months of isoniazid, rifampicin, pyrazinamide, and ethambutol followed by 4 months of isoniazid and rifampicin.	High	Strong
A 4-month regimen containing rifapentine, moxifloxacin, isoniazid, and pyrazinamide may be used in eligible adolescents and adults with drug-susceptible pulmonary TB where drugs, monitoring, and eligibility assessment are available.	Moderate	Conditional

Directly observed treatment alone should not be used as a coercive or stand-alone adherence strategy; differentiated, patient-centred adherence support is preferred.	Moderate	Strong
Fixed-dose combinations are preferred where quality-assured formulations are available to reduce pill burden and prescribing errors.	Moderate	Strong
Treatment response should be monitored clinically and microbiologically, with sputum testing where appropriate, and patients should be assessed for drug toxicity and adherence barriers throughout therapy.	Moderate	Strong

D. Drug-Resistant TB Treatment

Recommendation	GRADE Evidence Quality	Strength
All people diagnosed with rifampicin-resistant or multidrug-resistant TB should have rapid testing for fluoroquinolone resistance to guide regimen choice.	High	Strong
For eligible people with MDR/RR-TB without fluoroquinolone resistance, a 6-month all-oral regimen containing bedaquiline, pretomanid, linezolid, and moxifloxacin may be used.	Moderate	Conditional
For eligible people with MDR/RR-TB with additional resistance patterns or intolerance to moxifloxacin, a bedaquiline–pretomanid–linezolid-based regimen may be considered according to WHO criteria.	Moderate	Conditional
Injectable agents should generally be avoided when effective all-oral regimens are available.	Moderate	Strong
Longer individualized regimens should be constructed using WHO drug grouping and resistance profile when short regimens are unsuitable.	Moderate	Strong
Active drug-safety monitoring and management should be implemented for all people receiving second-line TB medicines, especially bedaquiline, linezolid, clofazimine, fluoroquinolones, and pretomanid.	Moderate	Strong

E. Children and Adolescents

Recommendation	GRADE Evidence Quality	Strength
Children and adolescents with signs or symptoms of TB should receive rapid molecular testing as the initial diagnostic test using appropriate respiratory or non-respiratory specimens.	Moderate	Strong
Children with non-severe, presumed drug-susceptible TB may receive a 4-month treatment regimen when WHO eligibility criteria are met.	Moderate	Strong

Children and adolescents with severe pulmonary TB, disseminated TB, osteoarticular TB, or central nervous system TB require longer or disease-specific treatment according to WHO guidance.	Moderate	Strong
Child-friendly fixed-dose combinations should be used where available.	Moderate	Strong
Household child contacts of people with bacteriologically confirmed pulmonary TB should be screened and offered TB preventive treatment if eligible after active TB is excluded.	High	Strong
Adolescents should be included in age-appropriate adherence, mental-health, nutrition, sexual-health, and social-support services.	Low	Conditional

1.4 Guideline Development Methodology

• Evidence Review Process

- WHO used systematic evidence reviews based on predefined PICO questions.
- Evidence sources included randomized controlled trials, diagnostic accuracy studies, cohort studies, individual patient data meta-analyses, systematic reviews, programme data, and pharmacovigilance data.
- Outcomes prioritized included mortality, cure, treatment success, relapse, treatment failure, adverse events, diagnostic accuracy, time to diagnosis, loss to follow-up, resistance amplification, acceptability, feasibility, equity, and cost.

• GRADE Framework

- Certainty of evidence was assessed as **High, Moderate, Low, or Very Low**.
- Recommendations were classified as **Strong or Conditional**.
- Decisions considered certainty of evidence, balance of benefits and harms, values and preferences, resource implications, equity, acceptability, and feasibility.

• Consensus Method

- WHO Guideline Development Groups reviewed evidence-to-decision frameworks.
- Recommendations were developed through structured discussion and consensus.
- When consensus was not immediately reached, further deliberation was undertaken.

• Conflict of Interest Management

- Guideline Development Group members declared financial, professional, academic, and other interests.
- WHO reviewed declarations before participation.
- Members with relevant conflicts were restricted or excluded from decision-making where necessary.

• Youth Engagement Note

- The prompt-identified title on **meaningful youth engagement** is best interpreted as a WHO methodological case study rather than a separate clinical TB guideline.
- Its relevance is strongest for adolescent TB guideline development, particularly in

ensuring that recommendations reflect adolescent acceptability, feasibility, stigma, adherence barriers, and rights-based care.

II. Evidence Summary After WHO Latest Guideline Release (WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities, 2nd ed.) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Because the report's specified WHO latest-update date and evidence cutoff date are both **29 April 2026**, there is no post-cutoff interval after that date. Therefore, no evidence can be considered newer than the stated cutoff. The following section summarizes the evidence surveillance approach and the key evidence base that supports or contextualizes the current WHO recommendations.

Search Strategy

- **Date range for surveillance:** From release of relevant WHO module updates through 29 April 2026.
- **Core sources searched or prioritized conceptually:**
 - WHO guideline repository and WHO Global Tuberculosis Programme documents
 - PubMed/MEDLINE
 - Cochrane Database of Systematic Reviews
 - ClinicalTrials.gov
 - International Clinical Trials Registry Platform
 - CDC TB guidance
 - ECDC TB surveillance and guidance
 - NIH HIV opportunistic infection guidelines
 - Major journals: *New England Journal of Medicine*, *The Lancet*, *JAMA*, *BMJ*, *Clinical Infectious Diseases*

Inclusion Criteria

- Randomized controlled trials, cluster trials, diagnostic accuracy studies, systematic reviews, meta-analyses, individual patient data meta-analyses, and major implementation studies.
- Studies addressing TB diagnosis, drug-susceptible TB treatment, drug-resistant TB treatment, paediatric TB, TB/HIV, TB/diabetes, nutrition, adherence, comorbidities, or service integration.
- Studies with clinically relevant outcomes: mortality, cure, relapse, treatment success, microbiological confirmation, adverse events, transmission, adherence, loss to follow-up, and patient-important outcomes.

Exclusion Criteria

- Non-systematic narrative reviews without new data.

- Small case series not affecting guideline certainty.
- Studies using obsolete regimens or diagnostics without relevance to current WHO practice.
- Preclinical studies without clinical outcome data.

2.2 Key Evidence Base and Practice-Relevant Studies

Study / Source	Design	Key Findings	GRADE Quality
Dorman SE et al., <i>New England Journal of Medicine</i> , 2021	Phase 3 randomized trial	A 4-month rifapentine–moxifloxacin regimen was non-inferior to standard 6-month therapy in eligible people with drug-susceptible pulmonary TB.	High
Turkova A et al., <i>New England Journal of Medicine</i> , 2022	Randomized non-inferiority trial	A 4-month regimen was non-inferior to 6 months of treatment in children with non-severe drug-susceptible TB.	High
Nyang’wa BT et al., <i>New England Journal of Medicine</i> , 2022	Randomized controlled trial	A 24-week all-oral regimen for rifampicin-resistant TB improved treatment outcomes compared with standard care in eligible patients.	Moderate
Conradie F et al., <i>New England Journal of Medicine</i> , 2022	Randomized dose-optimization trial	Bedaquiline–pretomanid–linezolid regimens showed high efficacy in highly drug-resistant TB, with linezolid toxicity influenced by dose and duration.	Moderate
Paton NI et al., <i>New England Journal of Medicine</i> , 2023	Adaptive strategy trial	Treatment-shortening strategies for selected rifampicin-susceptible TB populations showed potential but require careful selection and monitoring before broad guideline adoption.	Moderate
WHO rapid diagnostic evidence reviews supporting Module 3	Systematic reviews and diagnostic accuracy evidence	Rapid molecular diagnostics improve early detection of TB and rifampicin resistance compared with smear microscopy, especially when used as the initial test.	High
WHO LF-LAM evidence reviews	Systematic review / diagnostic accuracy evidence	LF-LAM improves TB diagnosis in selected people living with HIV, especially those hospitalized, seriously ill, or with advanced immunosuppression.	Moderate
WHO drug-resistant TB evidence reviews	Individual patient data meta-analysis and trial evidence	All-oral shorter regimens reduce treatment burden and avoid injectable-related toxicity; regimen choice depends on resistance profile and eligibility.	Moderate
WHO child and adolescent TB evidence reviews	Systematic reviews and paediatric trials	Shorter regimens are effective for non-severe paediatric TB; child-friendly diagnostics and formulations improve feasibility and acceptability.	Moderate

WHO TB/HIV and comorbidity evidence reviews	Systematic reviews and guideline synthesis	Early ART, TPT, integrated HIV/TB care, nutritional support, and comorbidity management reduce mortality and improve outcomes.	High to Moderate
---	--	--	------------------

2.3 GRADE Assessment of New or Practice-Relevant Evidence

Evidence Domain	Benefits	Harms / Limitations	Certainty of Evidence	Recommendation Implication
Rapid molecular testing as initial TB diagnostic test	Earlier diagnosis, earlier rifampicin-resistance detection, faster treatment initiation	Cost, infrastructure, cartridge supply, need for quality assurance	High	Continue strong recommendation
LF-LAM in eligible people living with HIV	Mortality benefit in high-risk hospitalized or advanced-HIV groups when linked to treatment	Lower sensitivity in lower-risk groups; adjunctive test only	Moderate	Continue strong targeted recommendation
4-month rifapentine–moxifloxacin regimen for drug-susceptible TB	Shorter therapy, potential adherence benefit	Requires eligibility assessment, drug availability, drug–drug interaction review	Moderate	Conditional use in to High eligible patients
4-month regimen for non-severe paediatric TB	Less treatment burden, non-inferior outcomes in selected children	Requires accurate classification of non-severe disease	High	Strong recommendation for eligible children
6-month BPaLM/BPaL-based regimens for MDR/RR-TB	Shorter, all-oral treatment; improved feasibility and reduced injectable toxicity	Linezolid toxicity, QT monitoring, resistance concerns, pregnancy and paediatric limitations depending on setting	Moderate	Conditional use in eligible patients
Integrated TB/HIV care	Reduced mortality, improved uptake of ART and TPT	Implementation complexity	High	Strong recommendation
TB and diabetes bidirectional screening	Earlier detection of comorbidity; potential improved outcomes	Evidence for outcome improvement is less direct; resource burden	Low	Conditional targeted implementation

Nutrition assessment and support	Addresses major risk factor for mortality and poor treatment outcome	Optimal intervention packages vary by setting	Moderate	Strong recommendation for assessment and support
Mental health and substance-use integration	May improve adherence and retention	Evidence heterogeneous; implementation gaps	Low	Conditional but clinically important

2.4 Evidence Gaps and Future Research Needs

1. TB and diabetes

- Optimal screening frequency and cost-effective models for bidirectional TB–diabetes care.
- Best glucose-management strategies during TB treatment.
- Impact of diabetes control on relapse, mortality, and transmission.

2. TB and undernutrition

- Which nutritional interventions most improve survival and treatment success.
- Optimal packages for household food insecurity and social protection.

3. TB and mental health

- Effectiveness of depression, anxiety, alcohol-use, and substance-use interventions integrated into TB care.
- Scalable models in high-burden, low-resource settings.

4. Drug-resistant TB

- Safety and effectiveness of BPaLM/BPaL-based regimens in pregnant people, children, extrapulmonary TB, severe disease, and people with extensive prior second-line drug exposure.
- Best approaches to linezolid dose, duration, toxicity monitoring, and management.
- Resistance emergence after shorter regimens.

5. Diagnostics

- Performance of rapid molecular tests using non-sputum specimens.
- Affordable decentralized drug-susceptibility testing.
- Diagnostic algorithms for smear-negative, extrapulmonary, paediatric, and HIV-associated TB.

6. Children and adolescents

- Better diagnostics for paucibacillary disease.
- Adolescent-specific adherence and mental-health interventions.
- Inclusion of youth preferences in guideline development and implementation.

7. Implementation science

- How best to integrate TB, HIV, diabetes, nutrition, mental health, and social protection without overburdening health systems.
- Digital adherence technologies, community-based models, and differentiated care.

2.5 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact by 29 April 2026	Updated Recommendation
Use WHO-recommended rapid molecular tests as the initial diagnostic test for TB and rifampicin resistance.	Evidence remains strongly supportive.	Maintain strong recommendation. Expand access, decentralization, quality assurance, and linkage to treatment.
Use LF-LAM in eligible people living with HIV.	Evidence remains supportive for high-risk HIV populations.	Maintain targeted strong recommendation; do not use as a stand-alone rule-out test.
Treat most drug-susceptible TB with 6-month rifampicin-based therapy.	Shorter regimens are effective in selected groups but not universal.	Maintain 6-month regimen as broadly applicable; use 4-month rifapentine–moxifloxacin regimen where eligibility, drugs, and monitoring are available.
Use 4-month treatment for non-severe paediatric drug-susceptible TB.	Evidence supports non-inferiority in carefully selected children.	Maintain strong recommendation for eligible children; ensure accurate severity classification.
Use all-oral shorter regimens for eligible MDR/RR-TB.	Trial evidence supports shorter all-oral regimens but safety and resistance monitoring remain essential.	Maintain conditional recommendation for BPaLM/BPaL-based regimens in eligible patients; ensure DST, ECG and toxicity monitoring, and pharmacovigilance.
Integrate TB and HIV services, including early ART and TPT.	Mortality benefit remains clear.	Maintain strong recommendation; prioritize same-site or closely linked TB/HIV services.
Screen and manage TB-associated comorbidities such as diabetes, undernutrition, mental health conditions, and substance use.	Evidence supports clinical importance; implementation evidence remains variable.	Maintain targeted screening and integrated care; adapt to local epidemiology and resources.
Provide nutritional assessment and support for people with TB.	Undernutrition remains a major predictor of mortality and poor outcomes.	Maintain strong recommendation; link TB care to nutrition and social-protection services.

References

1. World Health Organization. *WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities, 2nd ed.* Geneva: World Health Organization; 2024.
2. World Health Organization. *WHO consolidated guidelines on tuberculosis: module 3:*

diagnosis: rapid diagnostics for tuberculosis detection. Geneva: World Health Organization; 2024.

3. World Health Organization. *WHO consolidated guidelines on tuberculosis: module 4: treatment: drug-susceptible tuberculosis treatment*. Geneva: World Health Organization; 2022.
4. World Health Organization. *WHO consolidated guidelines on tuberculosis: module 4: treatment: drug-resistant tuberculosis treatment, 2022 update*. Geneva: World Health Organization; 2022.
5. World Health Organization. *WHO consolidated guidelines on tuberculosis: module 5: management of tuberculosis in children and adolescents*. Geneva: World Health Organization; 2022.
6. Dorman SE, Nahid P, Kurbatova EV, et al. Four-month rifapentine regimens with or without moxifloxacin for tuberculosis. *N Engl J Med*. 2021;384:1705-1718.
7. Turkova A, Wills GH, Wobudeya E, et al. Shorter treatment for nonsevere tuberculosis in African and Indian children. *N Engl J Med*. 2022;386:911-922.
8. Nyang'wa BT, Berry C, Kazounis E, et al. A 24-week, all-oral regimen for rifampin-resistant tuberculosis. *N Engl J Med*. 2022;387:2331-2343.
9. Conradie F, Bagdasaryan TR, Borisov S, et al. Bedaquiline–pretomanid–linezolid regimens for drug-resistant tuberculosis. *N Engl J Med*. 2022;387:810-823.
10. Paton NI, Cousins C, Suresh C, et al. Treatment strategy for rifampin-susceptible tuberculosis. *N Engl J Med*. 2023;388:873-887.
11. Centers for Disease Control and Prevention. Tuberculosis clinical care and treatment guidance. Atlanta: CDC.
12. European Centre for Disease Prevention and Control. Tuberculosis surveillance and monitoring in Europe. Stockholm: ECDC.
13. National Institutes of Health. Guidelines for the prevention and treatment of opportunistic infections in adults and adolescents with HIV: tuberculosis section. Bethesda: NIH.